模集笔记

笔记面向复旦881考研应试，均为备考过程中记录，可能存在错误，希望大家仔细辨别，也欢迎大家指正。

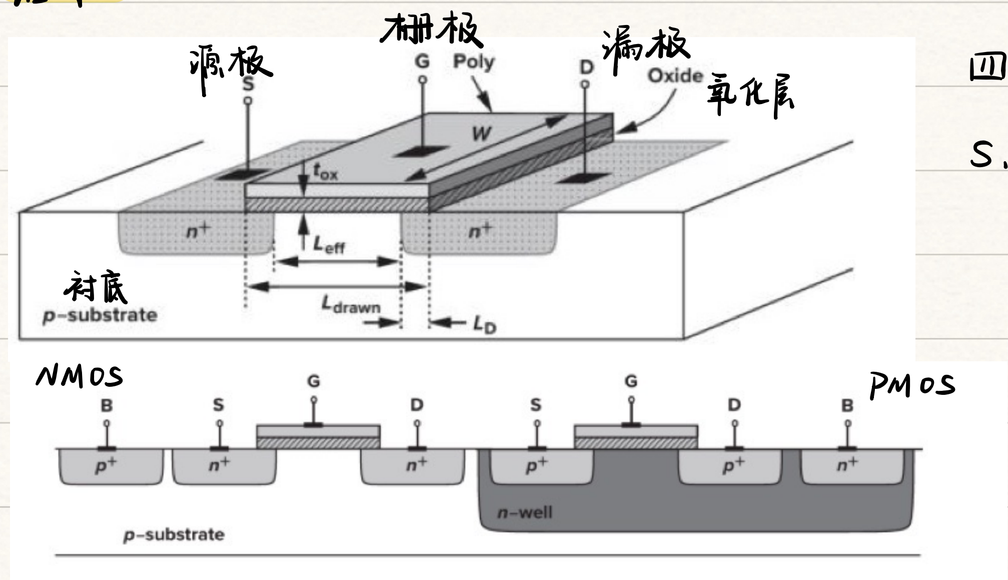
最后，恳请大家不要在其他平台散布这份笔记，感谢！

如果有考研规划，复试等其他问题，可以添加微信 yuzz9631

祝各位都能金榜题名，一战成硕！

§1. MOS 管

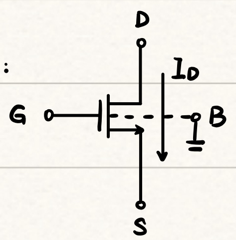
结构



四端器件（G、D、S、B）

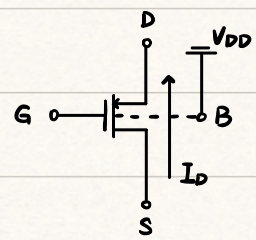
S、D对称

NMOS：



$V_D > V_S$

PMOS：



$V_S > V_D$

（B端为默认电位）

另：

题目中符号不标箭头时，看 V_D 和 V_S 关系判断 NMOS / PMOS

工作区域

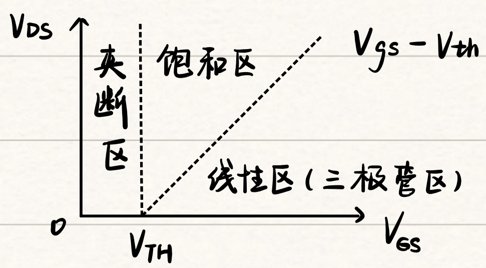
（以 NMOS 为例）

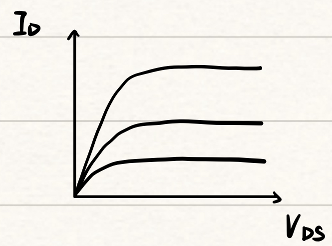
过驱动电压：$V_{GT} = V_{GS} - V_{TH}$

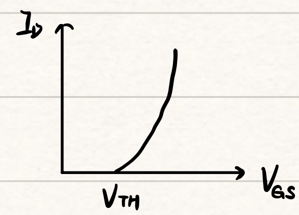
截止区：$I_D = 0$ （$V_{GT} < 0$）

线性区：$I_D = \mu_n C_{ox} \frac{W}{L} [(V_{GS}-V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2]$ （$V_{GT} > 0$，$V_{DS} < V_{GT}$）

饱和区：$I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{GS}-V_{TH})^2$ （$V_{GT} > 0$，$V_{DS} > V_{GT}$）→ 工作区域







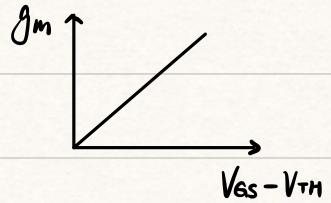
跨导

MOSFET工作在饱和区时：

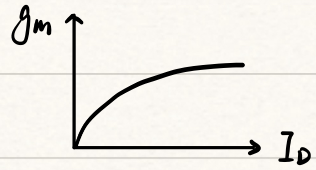
$$g_m = \frac{\partial I_D}{\partial V_{GS}}\Big|_{V_{DS}=constant}$$

三个公式等价，根据题中条件选公式

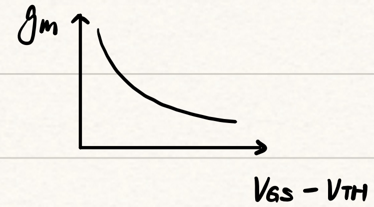
$$= \mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH}) \qquad = \sqrt{2\mu_n C_{ox}\frac{W}{L}I_D} \qquad = \frac{2I_D}{V_{GS}-V_{TH}}$$



$(\frac{W}{L}=constant)$



$(\frac{W}{L}=constant)$



$(I_D=constant)$

二级效应

1、体效应

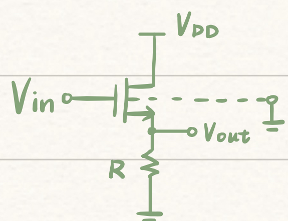
在NMOS中：$V_B<V_S$时，随V_B的下降，V_{th}增大，导致I_D减小

$$V_{TH} = V_{TH0} + \gamma\left(\sqrt{|2\Phi_F + V_{SB}|} - \sqrt{|2\Phi_F|}\right)$$

小信号图中：引入一个 $I = g_{mb}V_{bs}$ 的压控电流源

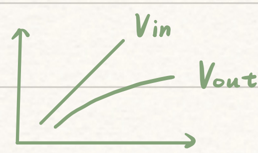
原理：

$V_{SB}>0$时，空穴被B区吸引，耗尽层变宽，需要更高的V_G形成反型层



$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{gs}-V_{th})^2 = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{in}-V_{out}-V_{th})^2$$

考虑体效应：$V_{in}\uparrow \rightarrow V_{out}\uparrow \rightarrow V_{SB}\uparrow \rightarrow V_{th}\uparrow$



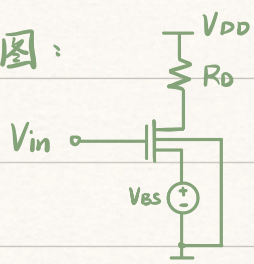
$$g_{mb} = \frac{\partial I_D}{\partial V_{BS}} = \mu_n C_{ox}\frac{W}{L}(V_{gs}-V_{th})\cdot\left(-\frac{\partial V_{th}}{\partial V_{BS}}\right) = \frac{\gamma}{2\sqrt{|2\Phi_F+V_{SB}|}}\cdot g_m = \eta g_m$$

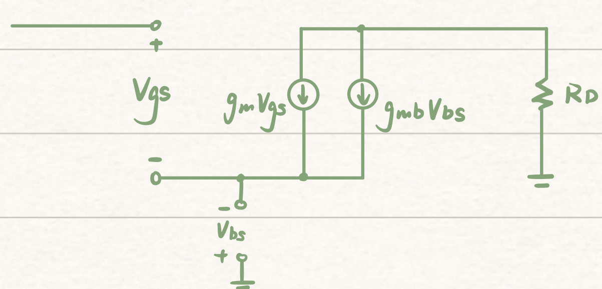
$\downarrow$

体效应系数

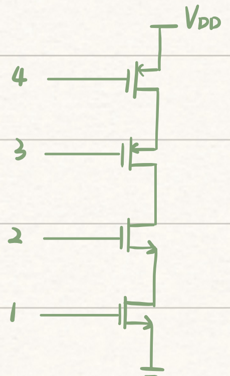
典型值：0.25

小信号图：





eg.



问：哪些管子中存在体效应？

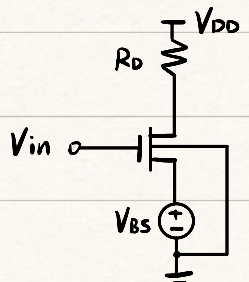
答：2、3管中存在体效应（$V_{SB} \neq 0$）

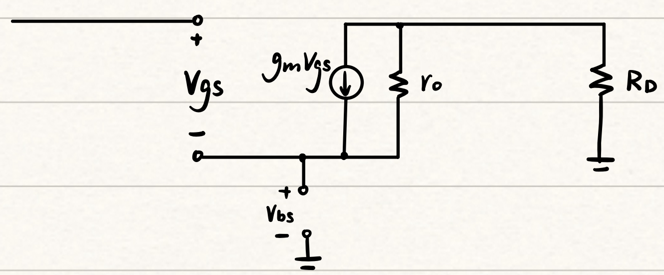
2、沟道长度调制效应

λ：沟道长度调制系数，$\lambda \propto \frac{1}{L}$，该效应只存在于饱和区

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2(1+\lambda V_{DS})$$

$$r_o = \frac{\partial V_{DS}}{\partial I_D} = \frac{1}{\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2\cdot\lambda} \approx \frac{1}{\lambda I_D} \propto \frac{L}{I_D}$$





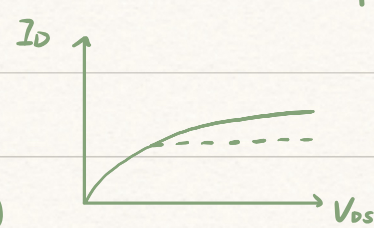
原理：

S→D：电压：$V_S \to V_D$，即 $V(x) = \frac{x}{L}\cdot V_{DS}$ 当 $V(x) = V_{gs}-V_{th}$ 时沟道夹断

有效沟道长度：$x = L\cdot\frac{V_{gs}-V_{th}}{V_{DS}} = L'$

$L' = L-\Delta L \to \frac{1}{L'} = \frac{1}{L-\Delta L} \approx \frac{1}{L}(1+\frac{\Delta L}{L}) \to$ 令 $\frac{\Delta L}{L} = \lambda V_{DS}$

$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L'}(V_{gs}-V_{th})^2 \approx \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{gs}-V_{th})^2(1+\lambda V_{DS})$



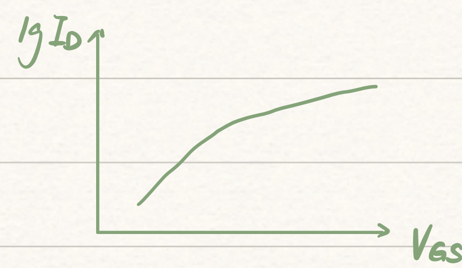
3、亚阈值导电性

当 V_{gs} 略小于 V_{th} 时，I_D 仍存在

$V_{DS} > 100mV$ 时，$I_D = I_0 e^{\frac{V_{gs}}{\xi V_T}}$

$$\begin{cases} I_0 \propto \frac{W}{L} \\ \xi > 1 \end{cases}$$

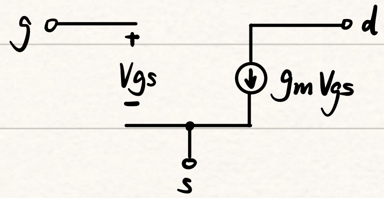
$V_T = \frac{kT}{q}$



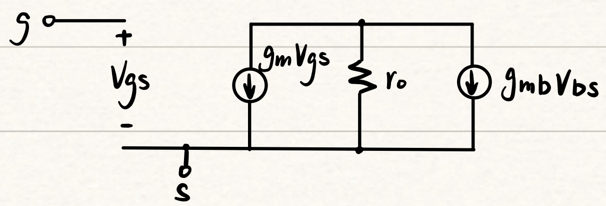
亚阈值区特性：功耗低、噪声大、速度慢

小信号图 (可以推导几乎所有电路，但过于复杂，因此一般用辅助定理)

不考虑二级效应：



考虑二级效应：



所有管子都有r_o。

$g_{mb} V_{bs}$是否存在需判断

注意：1) 栅端绝缘，无电流，可不画进小信号图，但要知道V_{GS}

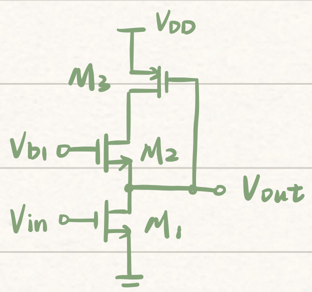
2) $g_m V_{GS}$以及$g_{mb} V_{BS}$的电流方向均为$d \to s$

3) 电路图中，除V_{in}、V_{out}外，写成V_{B1}、V_{B2}的节点电压都默认为恒定电位

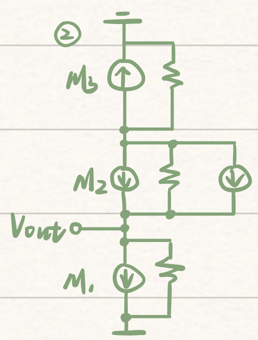
4) 在模集的小信号图中，电容一般视为断路

eg.1. 考虑r_o、g_{mb}画小信号图：

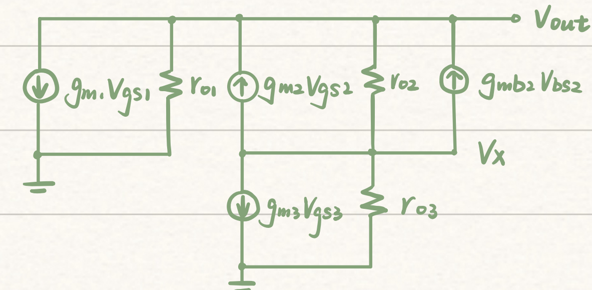
①判断：M_2中存在体效应



②



$\Rightarrow$ ③



列出A_v表达式

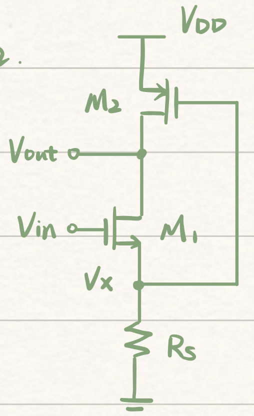
① 找V_{gs}：$V_{gs1} = V_{in}$，$V_{gs2} = -V_{out}$，$V_{gs3} = V_{out}$

V_{bs}：$V_{bs2} = -V_{out}$

② 列节点电流方程：$g_m V_{gs2} + \frac{V_x - V_{out}}{r_{o2}} + g_{mb2} V_{bs2} = g_{m3} V_{gs3} + \frac{V_x}{r_{o3}} = -g_{m1} V_{gs1} - \frac{V_{out}}{r_{o1}}$

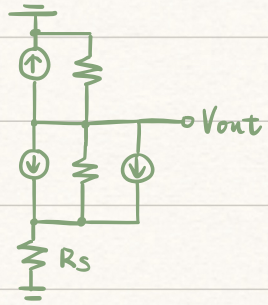
$$V_x = \left(\frac{r_{o1} r_{o3}}{r_{o1} + r_{o3}}\right)\left(g_{m1} V_{gs1} + g_{m3} V_{gs3}\right)$$

eg.2.



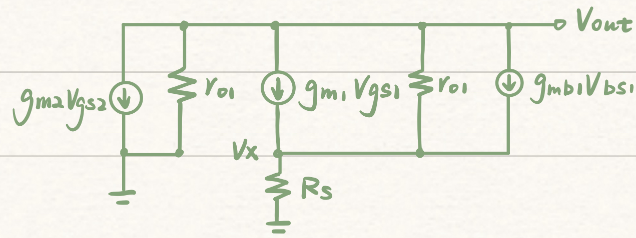
① 判断：M_1 中存在体效应

②



⇒

③



列 A_v 表达式：

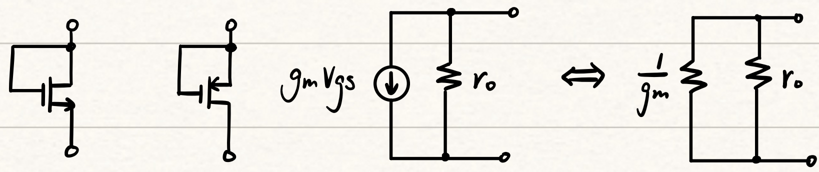
① $V_{gs1} = V_{in} - V_x \qquad V_{gs2} = V_x \qquad V_{bs1} = -V_x$

② $g_{m1}V_{gs1} + \dfrac{V_{out} - V_x}{r_{o1}} + g_{mb1}V_{bs1} = \dfrac{V_x}{R_s} = -g_{m2}V_{gs2} - \dfrac{V_{out}}{r_{o2}}$

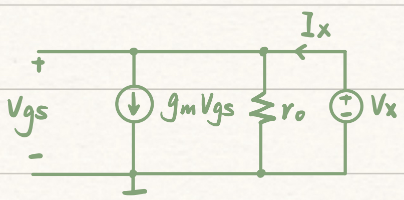
四种电路接法

等效电阻的求法：外加电压法（加独立源、其他电源置零）

1、栅漏短接（二极管连接型：起小信号电阻的作用）

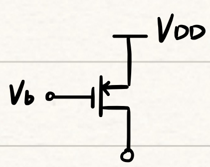


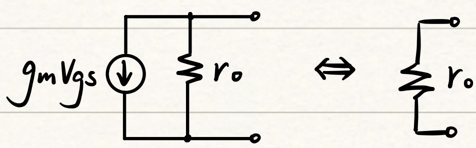
$R_{out} = \frac{1}{g_m} \,//\, r_o$



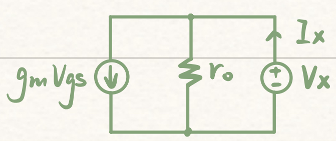
$R_{out} = \dfrac{V_x}{I_x} = \dfrac{V_{gs}}{g_mV_{gs} + \frac{V_{gs}}{r_o}} = \dfrac{1}{g_m + \frac{1}{r_o}} = \dfrac{1}{g_m} \,//\, r_o$

2、栅源短接（小信号图中短接）





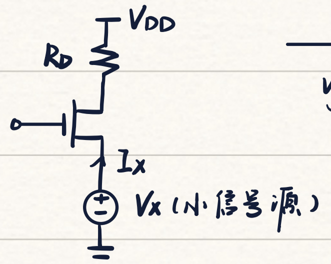
$R_{out} = r_o$

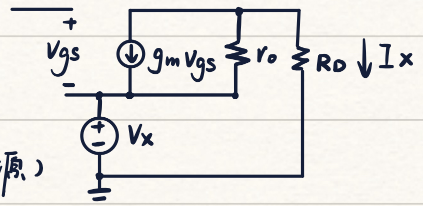


$V_{gs} = 0$

$I_x = \dfrac{V_x}{r_o} \Rightarrow R_{out} = \dfrac{V_x}{I_x} = r_o$

3、从源往漏看





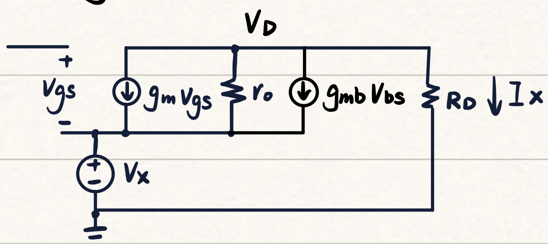
$$V_{gs} = -V_x$$

$$I_{r_o} = \frac{I_x R_D + V_x}{r_o} = -I_x - g_m V_{gs}$$

$$\Rightarrow I_x = \frac{(1 + r_o g_m) V_x}{R_D + r_o}$$

$$R_{out} = \frac{V_x}{I_x} = \frac{r_o + R_D}{1 + r_o g_m}$$

考虑 g_{mb} 时：

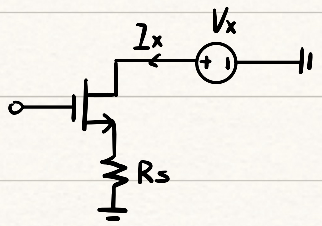


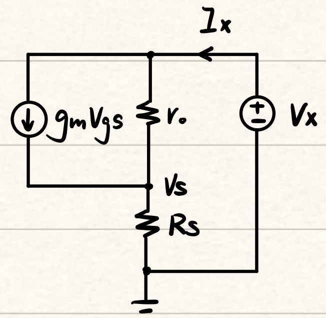
$$V_{gs} = -V_x \,, \quad V_{bs} = -V_x$$

$$I_x = \frac{V_D}{R_D} = -\frac{V_D - V_x}{r_o} - g_m V_{gs} - g_{mb} V_{bs}$$

$$R_{out} = \frac{V_x}{I_x} = \frac{R_D + r_o}{1 + (g_m + g_{mb}) r_o}$$

4、从漏往源看



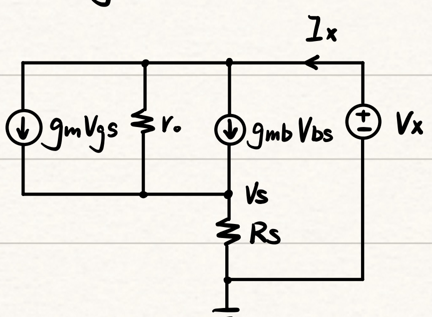


$$V_{gs} = -V_s$$

$$I_x = g_m V_{gs} + \frac{V_x - V_s}{r_o} = \frac{V_s}{R_s}$$

$$R_{out} = \frac{V_x}{I_x} = r_o + (1 + g_m r_o) R_s$$

考虑 g_{mb} 时：

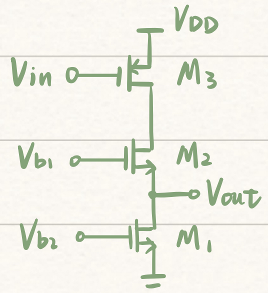


$$V_{gs} = -V_s \,, \quad V_{bs} = -V_s$$

$$I_x = g_m V_{gs} + g_{mb} V_{bs} + \frac{V_x - V_s}{r_o} = \frac{V_s}{R_s}$$

$$R_{out} = \frac{V_x}{I_x} = r_o + [1 + (g_m + g_{mb}) r_o] R_s$$

eg.3 求 Rout



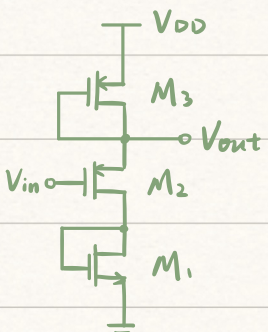
① 判断：M_2 中存在体效应

② $R_{out} = R_{out1} \,//\, R_{out2,3}$

$$= r_{o1} \,//\, \frac{r_{o2}+r_{o3}}{1+(g_m+g_{mb})r_{o2}}$$

（M_3 相当于栅源短接 → 看作 M_2 的 R_D）

eg.4 求 Rout

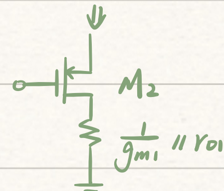


① 判断：M_2 中存在体效应

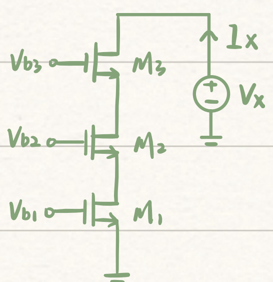
② $R_{out} = R_{out3} \,//\, R_{out1,2}$

$$= \frac{1}{g_{m3}} \,//\, r_{o3} \,//\, \frac{r_{o2} + \frac{1}{g_{m1}} \,//\, r_{o1}}{1+(g_{m2}+g_{mb2})r_{o2}}$$

$R_{out1,2}$：



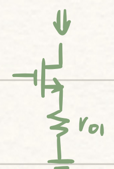
eg.4 求 Rout



① 判断：M_2、M_3 存在 g_{mb}

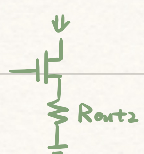
② $M_1 \Leftrightarrow r_{o1}$

M_2：



$\Leftrightarrow R_{out2} = r_{o2} + [1+(g_{m2}+g_{mb2})r_{o2}]\,r_{o1}$

M_3：

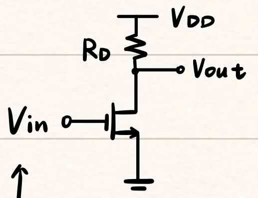


$$R_{out} = r_{o3} + [1+(g_{m3}+g_{mb3})r_{o3}]\{r_{o2}+[1+(g_{m2}+g_{mb2})r_{o2}]\,r_{o1}\}$$

§2. 放大器

判断方法同模电：既不做输入也不做输出的为公共端

共源放大器

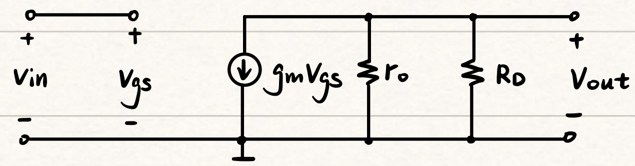


$V_{GG}+V_{in}$ (小信号)

静态工作点：

$$\begin{cases} V_{GSQ}=V_{GG} \\ I_{DQ}=\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2 \\ V_{DSQ}=V_{DD}-I_{DQ}R_D \end{cases}$$

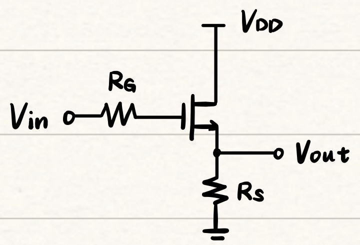
小信号图：



$$V_{gs}=V_{in}$$

$$A_v=\frac{-g_mV_{gs}(r_o\parallel R_D)}{V_{in}}=-g_m(r_o\parallel R_D)$$

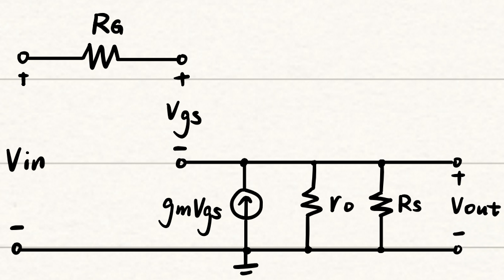
共漏放大器 (源跟随器)



静态工作点：

$$\begin{cases} I_{DQ}=\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2 \\ V_{GSQ}=V_{GG}-I_{DQ}R_S \\ V_{DSQ}=V_{DD}-I_{DQ}R_S \end{cases}$$

小信号图

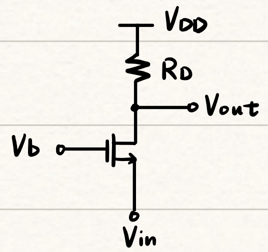


$$V_{gs}=V_{in}-V_{out}$$

$$V_{out}=g_mV_{gs}(r_o\parallel R_S)$$

$$A_v=\frac{V_{out}}{V_{in}}=\frac{g_mV_{gs}(r_o\parallel R_S)}{V_{gs}+g_mV_{gs}(r_o\parallel R_S)}=\frac{g_m(r_o\parallel R_S)}{1+g_m(r_o\parallel R_S)}$$

共栅放大器



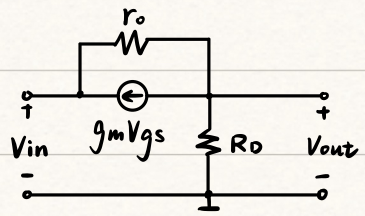
静态工作点

$$V_{GSQ} = V_b$$

$$I_{DQ} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2$$

$$V_{DSQ} = V_{DD} - I_{DQ}R_D$$

小信号图



$$V_{gs} = -V_{in}$$

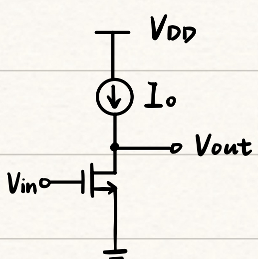
$$\frac{V_{out}}{R_D} + \frac{V_{out} - V_{in}}{r_o} + g_m V_{gs} = 0$$

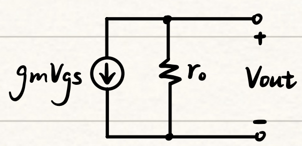
$$A_v = \frac{V_{out}}{V_{in}} = \frac{R_D + g_m R_D r_o}{R_D + r_o}$$

本征增益

1、定义：单极放大器所能达到的最大增益，即负载为无穷大的情况
（哪端输出就令哪端 $I=0$）不管电路如何改变，只要负载为无穷大
则本征增益保持不变

2、共源

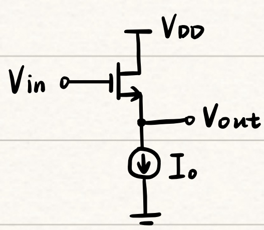


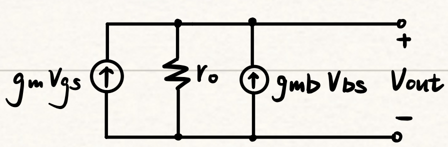


$$V_{gs} = V_{in}$$

$$A_v = -g_m r_o$$

3、共漏

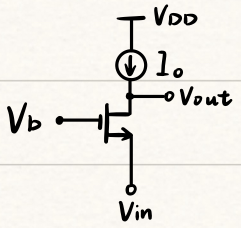


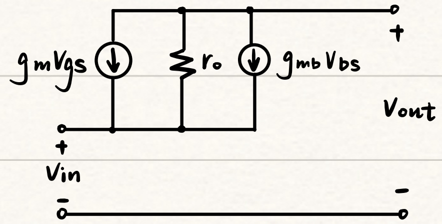


$V_{gs} = V_{in} - V_{out}$ ， $V_{bs} = -V_{out}$

$$A_v = \frac{g_m r_o}{1 + (g_m + g_{mb})r_o}$$ （共源/共栅）

4. 共栅

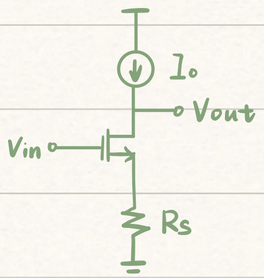


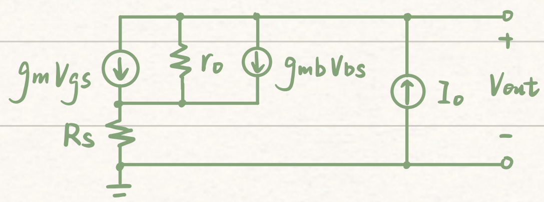


$V_{gs} = -V_{in}$ ， $V_{bs} = -V_{in}$

$A_v = 1 + (g_m + g_{mb}) r_o$

eg.1 证明A_v与R_s无关





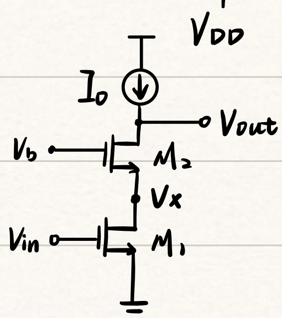
$I_o = 0 \Rightarrow$ 通过R_s的电流为0

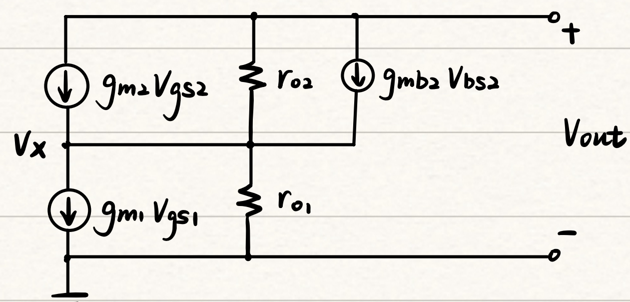
$\therefore V_{bs} = 0$

$\therefore A_v$与R_s无关

5. 共源共栅 (cascode)

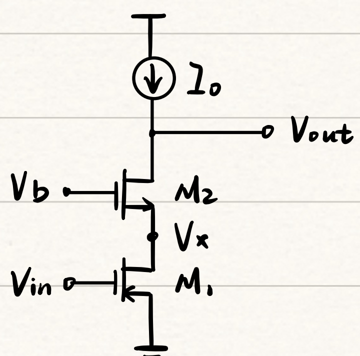
套筒式共源共栅





折叠式共源共栅

$V_{gs1} = V_{in}$ ， $V_{gs2} = -V_x$ ， $V_{bs2} = -V_x$



$A_v = -g_{m1} r_{o1} [1 + (g_{m2} + g_{mb2}) r_{o2}]$ (共源 × 共栅)

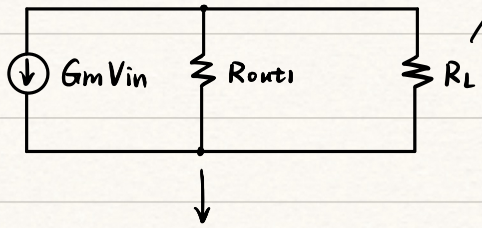
辅助定理

$G_m = g_m$ 时用这个公式

线性电路中，电压增益可等效为 $-G_m R_{out}$ ($R_{out} = R_{out1} \parallel R_L$)

G_m：输出与地短接时的跨导，$G_m = \frac{A_{v本}}{R_{out1}}$ → 符号由放大器类型判断

等效小信号图：



负载电阻（和 Vout 端相连的）

$$A_v = A_{v本} \cdot \frac{R_L}{R_{out} + R_L}$$

R_{out1}：除了输出端所接电阻外的电阻

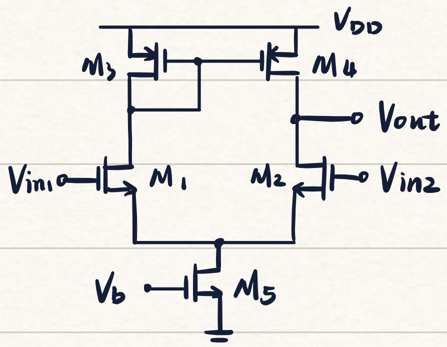
Tips：当共源放大器源极无电阻时，可直接认为等效跨导 $\approx g_m$

不考虑 r_{ds} 时，优先考虑用小信号图求解，也可令 $r_o \to \infty$

差分对

求解方法同模电，采用半边法

对于有源负载差分对 { 单边输出增益不减半
默认镜像电流源栅端接地

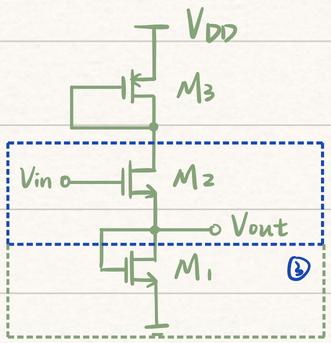


$G_m \approx g_{m2}$

$R_{out} \approx r_{o2} \parallel r_{o4}$

$A_v = g_{m2}(r_{o2} \parallel r_{o4})$

eg.2、计算A_v

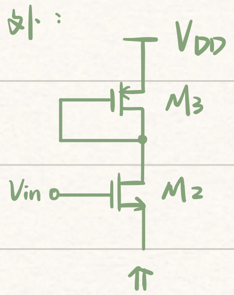


①判断：M_2中存在g_{mb}

②共漏：$A_{v本} = \dfrac{g_{m2} r_{o2}}{1+(g_{m2}+g_{mb2}) r_{o2}}$

③栅漏短接：$R_L = \dfrac{1}{g_{m1}} // r_{o1}$

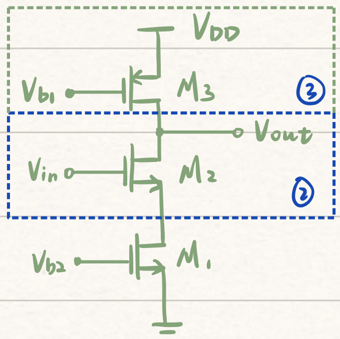
④除R_L外：



$$R_{out1} = \frac{(\frac{1}{g_{m3}} // r_{o3}) + r_{o2}}{1+(g_{m2}+g_{mb2}) r_{o2}}$$

⑤ $$A_v = A_{v本} \cdot \frac{R_L}{R_{out1}+R_L} = \frac{g_{m2} r_{o2} \cdot (\frac{1}{g_{m1}} // r_{o1})}{(\frac{1}{g_{m3}} // r_{o3}) + r_{o2} + [1+(g_{m2}+g_{mb2}) r_{o2}](\frac{1}{g_{m1}} // r_{o1})}$$

eg.3. 计算A_v

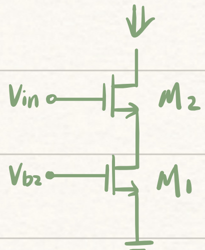


①判断：M_2中存在g_{mb}

②共源：$A_{v本} = -g_{m2} r_{o2}$

③栅漏短接：$R_L = r_{o3}$

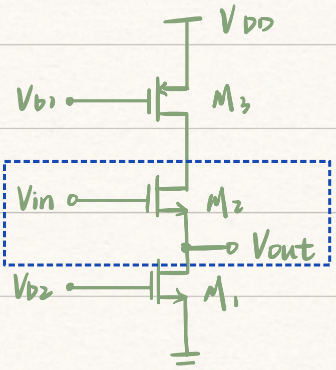
④除R_L外



$$R_{out1} = r_{o2} + [1+(g_{m2}+g_{mb2}) r_{o2}] r_{o1}$$

⑤ $$A_v = A_{v本} \cdot \frac{R_L}{R_{out1}+R_L} = -g_{m2} \cdot r_{o2} \cdot \frac{r_{o3}}{r_{o3}+r_{o2}+[1+(g_{m2}+g_{mb}) r_{o2}] r_{o1}}$$

eg.4 求Av

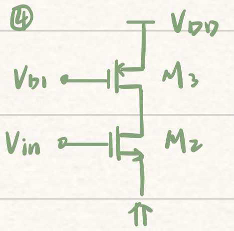


① 判断：M2中存在 gmb

② 共漏：$A_{v本}=\frac{g_{m2}r_{o2}}{1+(g_{m2}+g_{mb2})r_{o2}}$

③ R_L：栅源短接，$R_L=r_{o1}$

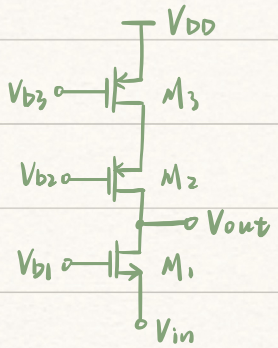
④



$R_{out1}=\frac{r_{o2}+r_{o3}}{1+(g_{m2}+g_{mb2})r_{o2}}$

⑤ $A_v=A_{v本}\cdot\frac{R_L}{R_{out1}+R_L}=\frac{g_{m2}r_{o1}r_{o2}}{r_{o2}+r_{o3}+[1+(g_{m2}+g_{mb2})r_{o2}]r_{o1}}$

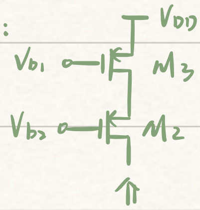
eg.5 求Av



① 判断：M1、M2中存在 gmb

② 共栅：$A_{v本}=1+(g_{m1}+g_{mb1})r_{o1}$

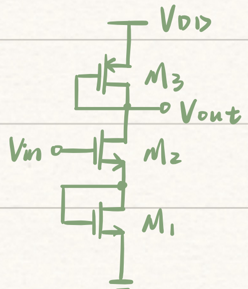
③ R_L：



$R_L=r_{o2}+[1+(g_{m2}+g_{mb2})r_{o2}]r_{o3}$

④ $R_{out1}=r_{o1}$

eg.6 求Av

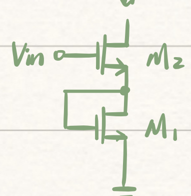


① 判断：M2中存在 gmb

② 共源：$A_{v本}=-g_{m1}r_{o1}$

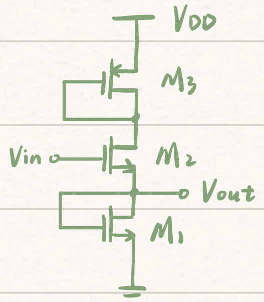
③ $R_L=\frac{1}{g_{m3}}//r_{o3}$

④



$R_{out1}=r_{o2}+[1+(g_{m2}+g_{mb2})r_{o2}](\frac{1}{g_{m1}}//r_{o1})$

eg.7 求 Av

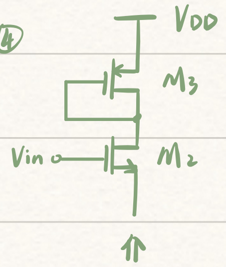


① 判断：M_2 中存在 g_{mb}

② 共漏：$A_{v本} = \dfrac{(g_{m2}+g_{mb2})r_{o2}}{1+(g_{m2}+g_{mb2})r_{o2}}$

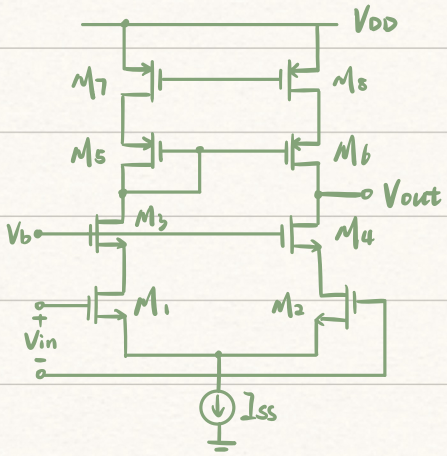
③ $R_L = \dfrac{1}{g_{m1}} \,//\, r_{o1}$

④



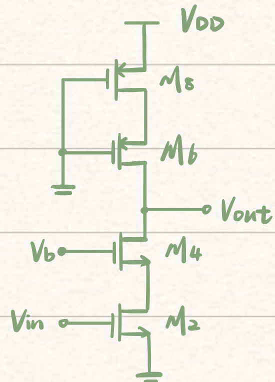
$$R_{out1} = \frac{r_{o2} + \left(\frac{1}{g_{m3}} // r_{o3}\right)}{1+(g_{m2}+g_{mb2})r_{o2}}$$

eg.8 求 Avd



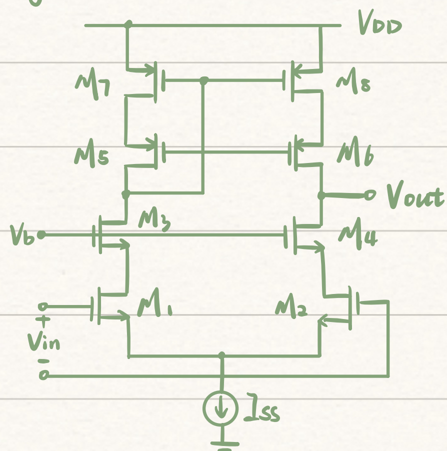
① $G_m \approx g_{m2}$

② Rout：



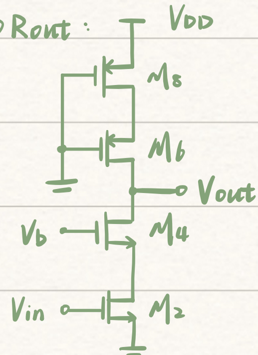
$$R_{out} = \{r_{o4} + [1+(g_{m4}+g_{mb4})r_{o4}]\,r_{o2}\} \,//\, \{r_{o6} + [1+(g_{m6}+g_{mb6})r_{o6}]\,r_{o8}\}$$

eg.9 求 Avd



① $G_m \approx g_{m2}$

② Rout：

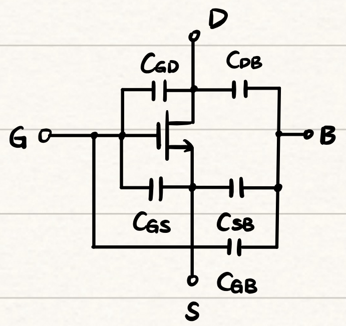


$$R_{out} = \{r_{o4} + [1+(g_{m4}+g_{mb4})r_{o4}]\,r_{o2}\} \,//\, \{r_{o6} + [1+(g_{m6}+g_{mb6})r_{o6}]\,r_{o8}\}$$

§3. 频率响应 (忽略 r_o)

MOS 器件电容

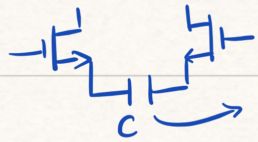
电容存在于 MOSFET 四端中的任意两端 (C_{SD} 忽略不计)

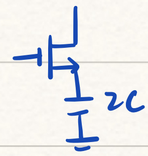


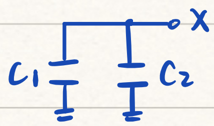
① 在饱和区不存在 C_{GB}

② $V_{SB}=0$ 时，不存在 C_{SB}

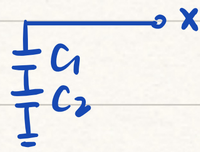
模集中的电容一般视为断开，电容并联为相加

若模中: 

画法: 

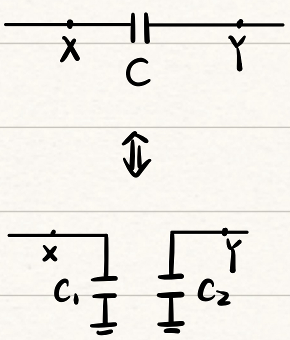


$\Rightarrow C_x = C_1 + C_2$



$\Rightarrow C_x = \dfrac{1}{\frac{1}{C_1}+\frac{1}{C_2}}$

密勒定理



$$\begin{cases} C_1 = (1-A_v)\cdot C \\ C_2 = (1-\frac{1}{A_v})\cdot C \end{cases}$$

应用条件: X和Y之间存在不只一个信号通路

$A_v < 0$

传输函数

模电中：

$$G(s) = \frac{V_{out}(s)}{V_{in}(s)}$$

$\left\{ \begin{array}{l} \text{极点：使 } A_v \to \infty \text{ 的频率点，即分母} = 0 \\ \text{零点：使 } A_v \to 0 \text{ 的频率点，即分子} = 0 \end{array} \right\}$ 计算时令实部 = 虚部

低通电路：$\frac{1}{1+SCR}$ → 包含1个极点

高通电路：$\frac{SCR}{1+SCR}$ → 包含1个极点，+1个零点

模集中：

零点，为令 $I_{out} = 0$ （$V_{out} = 0$）时的点

分离定理：

对于 $as^2 + bs + c = 0$ 的两个解：

$s_1 + s_2 = -\frac{b}{a}$ ， $s_1 s_2 = \frac{c}{a}$

两个极点分开得足够远，即 $s_2 \gg s_1$

$s_1 + s_2 \approx s_2 = -\frac{b}{a}$

$s_1 = \frac{c}{a s_2} = -\frac{c}{b}$